CHAPTER 9

NEURAL CONTROL AND COORDINATION

CHAPTER ESSENCE AT A GLANCE:
Electrochemical resting potentials, voltage-gated action potential propagation, and neurotransmitter synaptic transmission.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Resting Membrane Potential and Electrogenic Pumping:** The resting axonal membrane is polarized (-70 mV inside relative to exterior). Intracellular axoplasm has high K+ and low Na+ with impermeable negatively charged proteins; extracellular fluid has high Na+ and low K+. Maintained by the active electrogenic Na+/K+ ATPase pump, which transports 3 Na+ outwards for every 2 K+ inwards using 1 ATP.

• **Action Potential Depolarization and Repolarization:** Upon threshold stimulus, voltage-gated Na+ channels open, causing a rapid influx of Na+ into the axoplasm, reversing membrane polarity from -70 mV to +30 mV (depolarization). Within a millisecond, Na+ channels inactivate and voltage-gated K+ channels open, allowing rapid efflux of K+ restoring the negative resting potential (repolarization).

• **Chemical Synaptic Transmission Mechanisms:** At chemical synapses (synaptic cleft 20 nm), arrival of an action potential at the axon terminal opens voltage-gated Ca2+ channels; Ca2+ influx causes synaptic vesicles to fuse with the presynaptic membrane, releasing neurotransmitters (acetylcholine) via exocytosis into the cleft. Neurotransmitters bind specific receptors on the postsynaptic membrane, generating excitatory or inhibitory postsynaptic potentials.

• **Functional Anatomy of the Human Brain:** Forebrain: Cerebrum with cerebral cortex (sensory, motor, association areas coordinating complex functions), corpus callosum tract connecting hemispheres, Thalamus (sensory relay station), Hypothalamus (thermoregulation, hunger, thirst, neurohormone secretion, limbic system emotional behavior). Midbrain: Corpora quadrigemina (four optic lobes). Hindbrain: Cerebellum (convoluted surface for muscular balance and coordination), Pons (fiber tracts), and Medulla oblongata (regulates cardiovascular reflexes, respiration, and gastric secretions).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\mathrm{Na^+\text{-}K^+\ ATPase} = 3\mathrm{Na^+_{out}} \Longleftrightarrow 2\mathrm{K^+_{in}} \quad [\mathrm{Consumes\ 1\ ATP} \rightarrow \mathrm{Resting\ } -70\ \mathrm{mV}]$$

Electrogenic Ion Exchange Ratio of the Sodium-Potassium Pump

$$\mathrm{Action\ Potential} = \mathrm{Resting}\ (-70\ \mathrm{mV}) \xrightarrow{\mathrm{Na^+\ influx}} \mathrm{Depol}\ (+30\ \mathrm{mV}) \xrightarrow{\mathrm{K^+\ efflux}} \mathrm{Repol}\ (-70\ \mathrm{mV})$$

Voltage-Gated Axonal Membrane Potential Phases

$$\mathrm{Synaptic\ Sequence} = \mathrm{Action\ Potential} \rightarrow \mathrm{Ca^{2+}\ influx} \rightarrow \mathrm{ACh\ Exocytosis} \rightarrow \mathrm{Postsynaptic\ EPSP}$$

Neurotransmitter Release Cascade at Chemical Synapses

$$\mathrm{Brain\ Tripartite\ Axes} = \mathrm{Forebrain\ (Cerebrum,\ Hypothalamus)} + \mathrm{Midbrain} + \mathrm{Hindbrain\ (Cerebellum,\ Medulla)}$$

Primary Divisions of the Human Central Nervous System

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Sense Organs (Eye and Ear) and the Reflex Arc were completely deleted under the rationalized syllabus.

• **Alert:** The sodium-potassium pump transports 3 Na+ outwards for every 2 K+ inwards against their concentration gradients, consuming one ATP per cycle.

• **Alert:** The Hypothalamus contains centers controlling body temperature, hunger, thirst, and sleep, while the Medulla oblongata controls autonomic respiration, heartbeat, and gastric secretions.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Describe the ionic basis of the resting membrane potential in a neuron, highlighting the role of the Na+/K+ ATPase pump. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Resting polarization: Inside of the axon is negatively charged (-70 mV) relative to the outside because axoplasm contains high K+ and negatively charged proteins, while extracellular fluid contains high Na+. **[1 Mark]**

• Na+/K+ pump role: Actively transports 3 Na+ ions outwards for every 2 K+ ions inwards against concentration gradients consuming ATP, maintaining the electrogenic gradient. **[1 Mark]**

2. Distinguish between electrical synapses and chemical synapses with respect to the synaptic cleft and transmission speed. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Electrical synapses: Membranes of pre- and post-synaptic neurons are in very close physical contact via gap junctions; impulse transmission is instantaneous and extremely fast. **[1 Mark]**

• Chemical synapses: Separated by a fluid-filled synaptic cleft (about 20 nm); transmission depends on chemical neurotransmitters, resulting in a slight synaptic delay. **[1 Mark]**

3. Explain the sequence of ionic events responsible for generating an action potential (depolarization and repolarization) along an unmyelinated axon. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Stimulus and threshold: A threshold stimulus at a site increases membrane permeability to Na+ by opening voltage-gated sodium channels. **[1 Mark]**

• Depolarization: Rapid influx of Na+ reverses membrane polarity from -70 mV to +30 mV, creating the action potential or nerve impulse. **[1 Mark]**

• Repolarization: Voltage-gated Na+ channels close rapidly while voltage-gated K+ channels open, allowing K+ efflux out of the axon to restore the resting negative potential. **[1 Mark]**

4. Describe the molecular mechanism of synaptic transmission across a chemical synapse upon the arrival of an action potential at the axon terminal. [Score: 3 Marks]

Answer & Key Valuation Steps:

• When an action potential arrives at the axon terminal, it stimulates the opening of voltage-gated calcium channels, causing Ca2+ influx from the extracellular fluid. **[1 Mark]**

• Ca2+ induces synaptic vesicles to move toward and fuse with the presynaptic plasma membrane, releasing neurotransmitters (e.g., acetylcholine) into the synaptic cleft by exocytosis. **[1 Mark]**

• Neurotransmitter molecules diffuse across the cleft and bind to specific receptors on the postsynaptic membrane, opening ligand-gated ion channels and generating a new potential. **[1 Mark]**

5. State the anatomical location and primary physiological functions of the Hypothalamus and the Medulla oblongata in the human brain. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Hypothalamus: Located at the base of the thalamus in the forebrain; contains control centers regulating body temperature (thermostat), urge for eating and drinking, and secretes hypothalamic releasing and inhibiting neurohormones. **[1.5 Marks]**

• Medulla oblongata: Connected to the spinal cord in the hindbrain; contains vital autonomic reflex centers that control respiration, cardiovascular reflexes, gastric secretions, and swallowing. **[1.5 Marks]**

6. Explain the functions of the Corpus callosum and the Cerebellum in coordinating neurological activities in humans. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Corpus callosum: A large, prominent tract of transverse nerve fibers connecting the left and right cerebral hemispheres, facilitating interhemispheric communication and cognitive coordination. **[1.5 Marks]**

• Cerebellum: Possesses a highly convoluted surface providing additional space for numerous neurons; coordinates voluntary muscular movements and maintains bodily posture and balance. **[1.5 Marks]**

7. A neurotoxin selectively blocks voltage-gated calcium channels at presynaptic axon terminals without affecting voltage-gated sodium or potassium channels. Predict the effect on: (a) Action potential conduction along the axon, (b) Neurotransmitter exocytosis, and (c) Postsynaptic potential generation. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• (a) Action potential conduction: Conduction along the axon remains completely normal and unaffected because voltage-gated Na+ and K+ channels operate normally. **[1.5 Marks]**

• (b) Neurotransmitter exocytosis: Completely abolished because Ca2+ influx into the synaptic knob is blocked, preventing synaptic vesicle docking and exocytosis of acetylcholine. **[1.5 Marks]**

• (c) Postsynaptic potential generation: No postsynaptic potential (EPSP or IPSP) can be generated because no neurotransmitter binds to postsynaptic receptors, completely blocking synaptic transmission. **[1 Mark]**